

ModernMolBERT: A ModernBERT Encoder Family for SELFIES Molecular Language Modeling

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Abstract

We introduce MODERNMOLBERT, a Modern Molecular BERT family for small-molecule representation learning. MODERNMOLBERT adapts the ModernBERT encoder architecture to SELFIES molecular language modeling, using chemically aware tokenization to preserve molecular syntax and structure. We train base and large models on large-scale unlabeled molecular corpora and evaluate the resulting representations on standard molecular property prediction benchmarks. MODERNMOLBERT provides a compact, reproducible, encoder-only foundation for SELFIES-based chemical machine learning.

1 Introduction

ModernBERT is a Modern BERT [2] and APE means Atom Pair Encoding Leon et al. [1]. We need these for molecules.

2 Methods

Describe the methods clearly enough that the work can be reproduced.

3 Results

Describe the results.

Example table:

Table 1: Example table caption.

Condition	Estimate	Standard error
A	1.23	0.10
B	0.98	0.12

4 Discussion

Interpret the results, discuss limitations, and describe future work.

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Data and code availability

State where data and code are available, or explain any restrictions.

Author contributions

Describe author contributions here, if appropriate.

Competing interests

Declare competing interests, or write: The authors declare no competing interests.

References

- [1] Miguelangel Leon, Yuriy Perezhohin, Fernando Peres, Aleš Popovič, and Mauro Castelli. Comparing SMILES and SELFIES tokenization for enhanced chemical language modeling. 14(1): 25016. ISSN 2045-2322. doi: 10.1038/s41598-024-76440-8. URL <https://www.nature.com/articles/s41598-024-76440-8>.
- [2] Benjamin Warner, Antoine Chaffin, Benjamin Clavié, Orion Weller, Oskar Hallström, Said Taghadouini, Alexis Gallagher, Raja Biswas, Faisal Ladhak, Tom Aarsen, Nathan Cooper, Griffin Adams, Jeremy Howard, and Iacopo Poli. Smarter, Better, Faster, Longer: A Modern Bidirectional Encoder for Fast, Memory Efficient, and Long Context Finetuning and Inference.

A Supplementary methods

Additional methodological details can go here.

B Additional results

Additional figures, tables, or derivations can go here.